From Signatures to Signals: How AI Is Transforming Cyber Defense

A strategic briefing for leadership on AI-driven threat detection and threat intelligence

Ruthvik Nath Bandari

AAI6680: AI for Cybersecurity · Northeastern University · Mimoza Dimodugno, PhD

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EXECUTIVE SUMMARY

AI is now essential to cyber defense, but it needs oversight

- **Threats outpace signatures:** 82% of CrowdStrike's 2025 detections were malware-free.
- **AI pays off:** extensive security AI was linked to **$1.9M** lower breach costs and **80** fewer days.
- **Intelligence turns proactive:** Google predicted a SQLite exploit and cut it off first.
- **Attackers use AI too:** one actor compromised 600+ FortiGate devices using commercial AI.

RECOMMENDATION

Adopt AI detection and triage in phases.

Keep analysts approving high-impact actions.

Govern every AI tool (NIST CSF 2.0, AI RMF).

Sources: CrowdStrike (2026b); IBM (2025); Walker (2025); Krebs (2026); National Institute of Standards and Technology [NIST] (2023, 2024).

Three leaders, three AI techniques

Microsoft

GENERATIVE AI (LLM) TRIAGE

Use

LLM agents help analysts triage phishing and incidents.

Result

Trial: **77% better** verdict accuracy. Live SOCs: 30% shorter resolution time. Both are Microsoft studies.

CrowdStrike

BEHAVIORAL MACHINE LEARNING

Use

Models attacker behavior across endpoint, identity and cloud.

Result

MITRE 2025: **100% detection, zero false positives** (vendor's reading of an independent test).

Palo Alto Networks

INLINE DEEP LEARNING ON FILES

Use

Classifies files in real time, including files no signature covers yet.

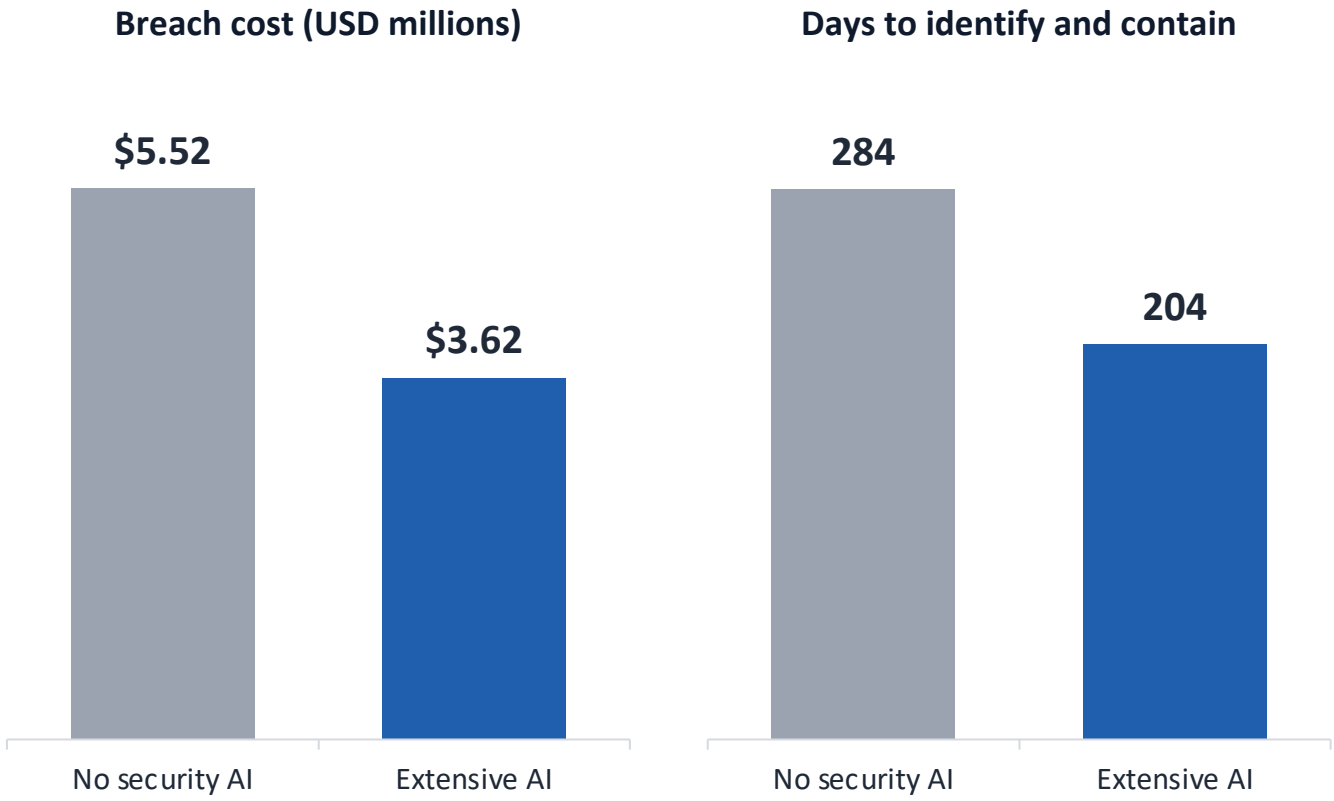
Result

Stops **99%+ of unknown malware** (vendor-reported).

Sources: Bono (2025); Bono et al. (2024); CrowdStrike (2025b); Palo Alto Networks (n.d.).

AI beats signatures on what is new, and the return is measurable

- **Signatures miss the new:** 23% of detected malware evaded them in late 2025 (vendor telemetry).
- **Detection improved:** a Fortune 500 AI pilot found threats 26–36% faster, with false positives down 16 points.
- **The savings held in 2026:** $1.93M lower cost per breach.



–$1.90M and –80 days with extensive security AI vs none (IBM, 2025). A correlation, not proof of cause.

Sources: WatchGuard Technologies (2026); Bracken (2026); IBM (2025, 2026).

What works best, and what gets in the way

MOST EFFECTIVE TECHNIQUES

- **Behavioral ML** for malware-free attacks
- **Deep learning** for unseen malware files
- **LLM assistants** for faster triage
- Best results: all three, plus **human review**

IMPLEMENTATION CHALLENGES

- **Noise:** 55% face too many false positives
- **Trust:** only 11% fully trust AI for critical tasks
- **Skills:** 59% report critical skill needs
- **AI as a target:** 97% of firms with AI breaches lacked AI access controls

Sources: Cisco (2025a); CrowdStrike (2026b); ISC2 (2025); IBM (2025).

AI threat intelligence at machine speed

Why speed matters: in 2025, average eCrime breakout time fell to **29 minutes**.

CrowdStrike

CHARLOTTE AI

SOC tool

Agentic alert triage, trained on millions of analyst decisions.

Result

98%+ triage accuracy and about 40 analyst hours saved a week (vendor).

SentinelOne

PURPLE AI

SOC tool

Agentic investigation, opened to all customers in 2026.

Result

MITRE 2024: all 16 attack steps detected, **88% fewer alerts** than the median (vendor).

Splunk (Cisco)

AI IN THE SIEM

SOC tool

Enterprise Security 8.2, with Triage and Malware Reversal agents announced alongside (Triage due 2026).

Industry survey

59% of security leaders say AI raised their efficiency.

Sources: CrowdStrike (2025a, 2026a); SentinelOne (2024, 2026); Cisco (2025a, 2025b).

From reactive to proactive threat intelligence

	Reactive (traditional)	Proactive (AI-driven)
Trigger	Known IOCs and signatures	Behavior and predicted exploits
Timing	After compromise	Before or during breakout
Hunting	Manual expert queries	Agentic investigation
Alerts	40% never investigated	AI agents triage and explain

EVIDENCE

- **Zero-day stopped:** Google's threat intelligence plus its Big Sleep AI predicted a SQLite exploit (CVE-2025-6965) and cut it off.
- **Early warning:** Darktrace flagged activity later traced to an Ivanti zero-day, 11 days before disclosure (vendor).
- **How well does AI predict?** Promising, but I found no independent study.

Sources: Townsend (2025); Walker (2025); Turnbull (2025).

Recommendation and reflection

CASE: AI-ASSISTED FORTIGATE CAMPAIGN (KREBS, 2026)

C: network maps and credentials exposed

I: firewall devices compromised by the attacker

A: at risk if those firewalls fail (my inference)

RECOMMENDATION

- **Protect:** MFA and no exposed management ports
- **Detect and respond:** behavioral AI with analyst approval
- **Govern:** access controls on every AI tool (AI RMF)

WHAT THIS MEANS FOR ME

My Module 1 and Module 3 detectors scored well, yet both missed attacks the headline numbers hid. I now read vendor claims the same way.

As a future security engineer, I want to build behavioral detectors with measured, explained limits, and a named analyst accountable for what they automate.

Sources: Krebs (2026); NIST (2023, 2024). Reflection draws on my Module 1 and Module 3 labs.

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